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COMMODITIES AND CAPABILITIES

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II
COMMODITIES AND THEIR USE

In an approach pioneered by Gorman (1956) and Lancaster (1966), commodities are seen in terms of their characteristics. The characteristics are the various desirable properties of the commodities in question. Securing amounts of these commodities gives the person command over the corresponding characteristics. For example, the possession of food gives the owner access to the properties of the food, which can be used to satisfy hunger, to yield nutrition, to give eating pleasure and to provide support for social meetings.[1]

However, the characteristics of the goods do not tell us what the person will be able to do with those properties. For example, if a person has a parasitic disease that makes the absorption of nutrients difficult, then that person may suffer from undernourishment even though he may consume the same amount of food as another person for whom that food is more than adequate.[2] In judging the well-being of the person, it would be premature to limit the analysis to the characteristics of goods possessed. We have to consider the 'functionings' of persons. While the ownership of commodities is a personal matter,[3] and thus the command over the characteristics of goods owned is also a personal matter, the quantification of characteristics does not vary with the personal features of the individual possessing the goods. A bicycle is treated as having the characteristic of 'transportation', and this is the case whether or not the particular person happening to possess the bike is able-bodied or crippled. In getting an idea of the well-being of the person, we clearly have to move on to 'functionings', to wit, what the person succeeds in *doing* with the commodities and characteristics at his or her command. For example,

[1] Scitovsky (1976), and Douglas and Isherwood (1979) have argued for taking a wider view of the use we make of goods than is found in mainline theory. See also Chipman, Richter and Sonnenschein (1971), Lancaster (1971), Gorman (1976), and Deaton and Muellbauer (1980).

[2] See Scrimshaw (1977). For the relevance of such variations in the perception of distributive justice, see Yaari and Bar-Hillel (1984, pp. 8–12).

[3] There are, of course, cases also of joint ownership and even of social ownership. In the case of such jointness, and also when the ownership is not joint but the *use* is meant to be (as with a family), there is a further problem of internal division of commodities commanded by the multiple-member unit. See Appendix B.

we must take note that a disabled person may not be able to do many things an able-bodied individual can, with the same bundle of commodities.

A functioning is an achievement of a person: what he or she manages to do or to be. It reflects, as it were, a part of the 'state' of that person. It has to be distinguished from the commodities which are used to achieve those functionings. For example, bicycling has to be distinguished from possessing a bike. It has to be distinguished also from the happiness generated by the functioning, for example, actually cycling around must not be identified with the pleasure obtained from that act. A functioning is thus different both from (1) having goods (and the corresponding characteristics), to which it is posterior, and (2) having utility (in the form of happiness resulting from that functioning), to which it is, in an important way, prior.

A little notation and specification might help. Consider

x_i = the vector of commodities possessed by person i,
$c(\cdot)$ the function (not necessarily linear) converting a commodity vector into a vector of characteristics of those commodities,
$f_i(\cdot)$ = a personal 'utilization function' of i reflecting one pattern of use of commodities that i can actually make (in generating a functioning vector out of a characteristic vector of commodities possessed),
F_i = the set of 'utilization functions' f_i, any one of which person i can in fact choose, and
$h_i(\cdot)$ = the happiness function of person i related to the functionings achieved by i.[4]

If the person chooses the utilization function $f_i(\cdot)$, then with his or her commodity vector x_i, the achieved functions will be given by the vector b_i,

$$b_i = f_i(c(x_i)). \tag{2.1}$$

The happiness that he will then enjoy is given by u_i,

$$u_i = h_i(f_i(c(x_i)). \tag{2.2}$$

[4] Here happiness is taken to be related to functionings only. It is easy to redefine it with other arguments as well, e.g., reflecting (say) the joys (if any) from possession. Note also that the functioning vector b_i may not depend only on the characteristics of the commodities possessed by the person himself (as assumed in equation 2.1), but may be influenced also by the functionings of others (e.g., illness may spread from one person from one person to another), public health and medical programmes, and so on.

If the person's choice of commodity vectors is restricted to set X_i, then the person's feasible functioning vectors are given by the set $Q_i(X_i)$,

$$Q_i(X_i) = [b_i \mid b_i = f_i(c(x_i)), \text{ for some } f_i(\cdot) \in F_i \text{ and for some } x_i \in X_i]. \tag{2.5}$$

$Q_i(X_i)$ represents the freedom that a person has in terms of the choice of functionings, given his personal features F_i (conversion of characteristics into functionings) and his command over commodities X_i ('entitlements').[6] Q_i can be called the 'capabilities' of person i given those parameters. It reflects the various combinations of functionings ('beings') he can achieve.

Given the valuation function $v_i(\cdot)$, it is of course possible to characterise the values of well-being that he can possibly achieve, given by the set V_i,

$$V_i = [v_i \mid v_i = v_i(b_i), \text{ for some } b_i \text{ in } Q_i]. \tag{2.6}$$

It must not be taken for granted that the highest value of v_i in V_i will necessarily be chosen (when such a maximum exists), since maximizing one's own well-being may not be the only motive for choice. Given other possible objectives and possible 'deontological' requirements (related, say, to one's obligations to others), it is quite possible that a non-v_i-maximal b_i may in fact be chosen. This will introduce an additional problem in the evaluation of a person's capabilities.

There is the further problem that in judging a 'freedom' type notion (as the concept of capabilities is), it is not altogether easy to identify the value of the set Q_i with the value of its highest-valued element (even when that element can and will be chosen). Consider a person having a capability set Q_i of which b_i^* is the unique maximal element (in terms of v_i), and assume first that b_i^* is in fact chosen, yielding well-being $v_i(b_i^*)$. Now assume that all b_i-vectors other than b_i^* become non-feasible (through a fall in entitlement X_i or in utilization F_i). But b_i^* is still achievable and $v_i(\cdot)$ is unchanged. The person's well-being will remain unaffected,[7] viz., $v_i(b_i^*)$, but it is not easy to claim that his 'freedom' is unchanged. In an important sense he can do less than he could do before, even though the best he can do is quite unchanged. These difficult issues will have to be discussed

[6] See Sen (1981a).

[7] Unless being able to choose is an important functioning and itself affects the person's well-being. This issue is taken up in Chapter 7.